

Bioinspired Clustering Algorithm for Credit Card User Segmentation Based on Mycelial Networks

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Abstract: Bioinspired optimization algorithms are increasingly applied in analyzing consumer behavior, particularly in financial datasets such as credit card transactions. This paper introduces the Fungi Kingdom Expansion (FKE) algorithm, a metaheuristic inspired by fungal growth, to optimize consumption patterns in large datasets. By simulating chaotic and deterministic fungal expansion, FKE effectively explores spending patterns, uncovering insights such as consumption trends and anomalies. In comparison to traditional methods like Particle Swarm Optimization (PSO), FKE excels in balancing local and global searches, making it highly suitable for optimizing credit card purchasing behaviors.

1 Introduction

In recent years, bioinspired algorithms have emerged as a crucial tool for solving complex optimization problems due to their effective balance between local and global search strategies [Lin et al., 2021]. The Fungi Kingdom Expansion (FKE) algorithm, inspired by the chaotic yet deterministic expansion of fungal mycelium, exemplifies this approach [Alnahwi et al., 2021]. Fungi expand their hyphae to find optimal conditions like moisture and temperature, and this adaptive strategy can be modeled computationally to efficiently explore search spaces [Fuchigami and Abreu, 2023].

The FKE algorithm employs a combination of chaotic (local) and deterministic (global) search mechanisms to optimize solutions iteratively [Alnahwi et al., 2021]. This approach mimics the immobile and mobile mass expansion observed in the fungal growth process [Zhao et al., 2022]. The immobile mass corresponds to the structure of the hyphae, while the mobile mass represents the flow of nutrients to the optimal tips of these hyphae [Huang et al., 2023]. Such a mechanism makes FKE particularly effective in avoiding local optima, a common problem in traditional optimization algorithms [Lin et al., 2021].

Fungi-inspired algorithms have demonstrated their potential in various engineering applications, such as antenna array optimization, where the FKE has shown superiority over traditional algorithms like Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) [Alnahwi et al., 2021]. These bioinspired approaches, in-

cluding Teaching-Learning-Based Optimization (TLBO), have been widely applied to solve large-scale industrial problems, including job shop and flow shop scheduling [Fuchigami and Abreu, 2023].

The adaptability of fungi-based models has opened new avenues in optimization, making them applicable to both single-objective and multi-objective optimization problems [Zhao et al., 2022]. For instance, the chaotic and random dispersion modes used by the FKE have proven to be effective in addressing complex problems where traditional algorithms may struggle to maintain a balance between exploration and exploitation [Lin et al., 2021].

1.1 State of the Art

The FKE algorithm replicates the chaotic and adaptive growth of fungal hyphae, leveraging local chaotic searches and global deterministic adjustments [Lin et al., 2021]. This duality enables effective navigation through complex search spaces, balancing exploration and exploitation [Zhao et al., 2022]. Applications include telecommunications, manufacturing, and financial data analysis, where FKE surpasses traditional algorithms in efficiency and adaptability [Huang et al., 2023]. By simulating evolutionary processes, FKE offers a biologically inspired advantage over conventional techniques [Fuchigami and Abreu, 2023].

1.2 Contextualization

Bioinspired computational techniques like FKE outperform traditional methods in high-dimensional and complex optimization scenarios [Zhao et al., 2022]. Fungi-inspired models adapt to environmental stimuli, emulating growth toward nutrient-rich regions. This adaptability enhances search efficiency and avoids premature convergence [Lin et al., 2021]. FKE's structured dual-phase—chaotic local exploration and directed global adjustment—provides a computational edge in diverse applications [Huang et al., 2023].

1.3 Key Methodologies

FKE's two-phase mechanism optimizes solutions iteratively. The chaotic local search explores the solution space broadly, while the deterministic global phase refines promising areas [Lin et al., 2021]. This iterative process avoids early convergence and enhances adaptability [Alnahwi et al., 2021]. Additionally, the spore dispersion mechanism enables exploration of distant solution areas when local options are insufficient [Fuchigami and Abreu, 2023].

1.4 Key Applications

FKE has been successfully applied in various fields:

1. Antenna Optimization: Enhanced radiation patterns with reduced interference compared to traditional algorithms [Alnahwi et al., 2021].
2. Anti-jamming Systems: Improved detection and neutralization of communication network interference [Fuchigami and Abreu, 2023].
3. Smart Manufacturing: Optimized resource allocation and scheduling, reducing operational costs [Zhao et al., 2022].
4. Data Center Energy Optimization: Minimized energy consumption in cooling systems [Huang et al., 2023].
5. Financial Data Analysis: Identified consumer behavior patterns and spending anomalies with higher precision [Lin et al., 2021].

1.5 Results and Advantages

FKE demonstrates notable advantages, including superior performance in large, complex search spaces and adaptability across diverse applications [Huang et al., 2023]. It outperforms traditional optimization methods in accuracy and speed, making it a valuable tool in engineering, manufacturing, and financial sectors [Fuchigami and Abreu, 2023].

1.6 Strategies Adopted by Other Authors

Researchers have explored several strategies for credit card user segmentation. Traditional methods like K-Means clustering have been utilized to group customers based on behavioral patterns, aiding in marketing and risk management [Barz et al., 2024]. Supervised algorithms such as Random Forest and XGBoost are frequently applied to predict

credit card approval, enhancing decision-making accuracy [Unirioja, 2024].

Bioinspired algorithms, which mimic natural processes, are also prominent for optimizing and analyzing financial datasets, offering insights into complex user behaviors [de Ingeniería del Conocimiento, 2024].

2 Problem Definition

The exponential growth in the use of credit cards has generated large volumes of transactional data containing valuable information about user behavior, from spending habits to consumption preferences. If properly analyzed, this data can help financial institutions make more informed decisions in areas such as fraud detection, credit risk management, and the personalization of offers and services. However, extracting value from these large datasets is not an easy task due to their complexity and the non-linear consumption patterns.

In this context, the objective of this project is to classify and segment banking users based on their use of credit cards. The proposed task is to address this segmentation using clustering techniques, a typical unsupervised learning method, since there are no predefined labels indicating which groups users belong to. Instead of using traditional clustering methods, this project proposes the development of a bioinspired algorithm based on fungal growth, emulating the mycelial expansion process to identify hidden patterns in the data.

The motivation for using a bioinspired approach lies in the unique adaptive growth characteristics that fungi exhibit in nature. The primary goal is to maximize the quality of classification through a process that follows biological principles, leveraging the decentralized and self-organized nature of mycelia. This approach could offer an advantage over traditional clustering algorithms by capturing the complex and stochastic patterns present in user data. However, the random and adaptive behavior of mycelial expansion presents challenges in terms of stability and consistency of the clusters, which could affect the results obtained.

The dataset used for this project includes approximately 8,950 user records, each described by 17 features detailing various aspects of credit card usage, such as account balance, purchase amounts, and credit limits, among other important characteristics. This dataset is representative of customers' financial behavior, and without a target variable, the task is to automatically uncover latent structures that enable the formation of coherent groups.

To evaluate the quality of the clustering, three standard metrics will be used: the Davies-Bouldin index, the silhouette score, and the Dunn index. The Davies-Bouldin index will assess the relationship between cluster compactness and separation, providing an effective measure of the relative separation of the groups. The silhouette score will measure how well points within a cluster are grouped compared to

others, offering insight into the proximity of points to their own clusters versus others. Finally, the Dunn index will help measure the distance between the closest clusters and the dispersion of the widest cluster, providing a reference for the separation between groups and their distribution in feature space.

An inherent limitation of the bioinspired approach is its stochastic behavior, which simulates the adaptive and decentralized properties of mycelial growth. This characteristic introduces a degree of variability in the results, which could affect the exact reproducibility of the clusters in each run of the algorithm. While this stochasticity can be a strength in terms of flexibility and adaptability to the data, it also presents a challenge when trying to ensure consistent results. The impact of this feature will be considered throughout the experiments and evaluations using the selected metrics.

In summary, this project aims to combine clustering techniques with bioinspired principles of fungal growth to identify usage patterns among banking users, enabling them to be grouped into clearly differentiated segments.

3 Dataset Presentation

In this project we will use the following dataset: **Credit Card Dataset for Clustering** from Kaggle [Bhasin, 2013]. This dataset contains information on more than 8,000 credit card users with 18 characteristics including total balance, frequency of purchases, credit limit, number of months the customer has had the card, among others.

It was found that there is a column with more than 300 missing values, which is `minimum_payment`, the minimum payment to be made by the customer. However, since the percentage of missing values in this column is very low relative to the total size of the dataset, we have decided to ignore it to simplify the analysis and avoid additional complications.

It is observed that there are variables with a strong correlation between each other, such as between the characteristics `purchases` and `oneoff_purchases`, see Figure 1:

4 Strategy for Solving the Problem

The strategy for developing and implementing the FKE focuses on several key components that will be essential to ensure that FKE provides a robust and effective solution for clustering credit card users. This strategy aims not only to improve the accuracy in identifying consumption patterns but also to provide an innovative tool capable of adapting to the complexities and changing dynamics of financial data.

The different stages of the strategy to be implemented are as follows:

Table 1: Statistical Summary of Variables

Variable	Count	Mean	Min	Max
BALANCE FREQUENCY	8950	0.88	0.00	1.00
PURCHASES	8950	1003.20	0.00	49039.57
ONEOFF PURCHASES	8950	592.44	0.00	40761.25
INSTALLMENTS PURCHASES	8950	411.07	0.00	22500.00
CASH ADVANCE	8950	978.87	0.00	47137.21
PURCHASES FREQUENCY	8950	0.49	0.00	1.00
ONEOFF PURCHASES FREQ	8950	0.20	0.00	1.00
PURCHASES INSTALL FREQ	8950	0.36	0.00	1.00
CASH ADVANCE FREQ	8950	0.14	0.00	1.50
CASH ADVANCE TRX	8950	3.25	0.00	123.00
PURCHASES TRX	8950	14.71	0.00	358.00
CREDIT LIMIT	8949	4494.45	50.00	30000.00
PAYMENTS	8637	1733.14	0.00	50721.48
MINIMUM PAYMENTS	8637	864.21	0.02	76406.21
PRC FULL PAYMENT	8950	0.15	0.00	1.00
TENURE	8950	11.52	6.00	12.00

1. **Data Analysis and Preparation:** The first fundamental step is data collection and preparation. Raw data must undergo preprocessing, including cleaning and normalization. This process is essential to ensure the quality and consistency of the data, which involves handling missing values, removing outliers, and normalizing variables to facilitate analysis. Next, a data exploration phase is conducted through exploratory analysis to better understand the characteristics of the data and detect preliminary patterns that can guide the development of the algorithm.
2. **Development of the Fungi Kingdom Expansion (FKE) Algorithm:** With the data prepared, the next step is the development of the Fungi Kingdom Expansion (FKE) algorithm. Inspired by the chaotic and deterministic expansion of fungal mycelium, the algorithm's design must reflect two main components: first, the stationary mass representing the structure of the hyphae that extends deterministically in search of resources; and second, the mobile mass that simulates the flow of nutrients to the optimal tips of the hyphae, promoting global exploration and adaptation. The algorithm must incorporate search mechanisms that combine chaotic (local) and deterministic (global) strategies to optimize user clustering. The implementation must consider how FKE can effectively balance local and global exploration to identify significant patterns in the data.
3. **Algorithm Configuration and Execution:** Once

5 Workflow

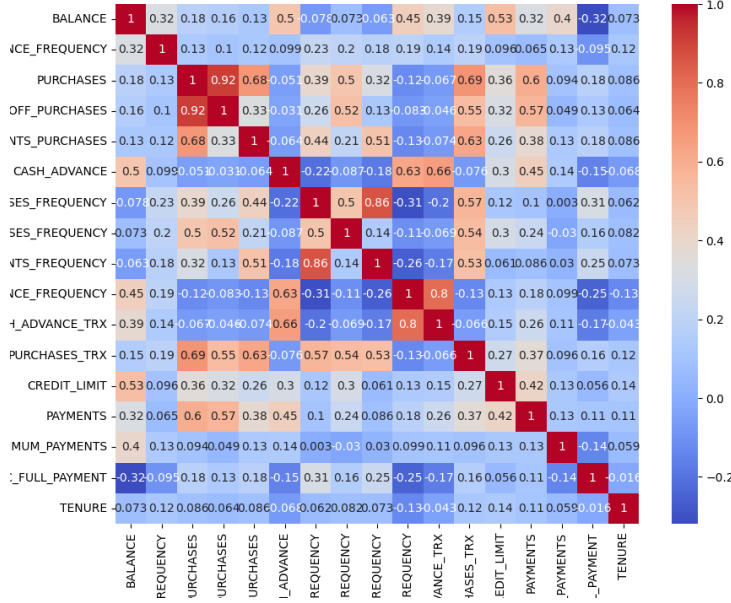
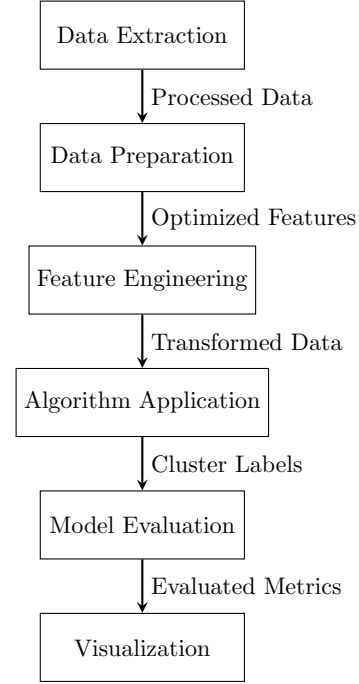


Figure 1: Correlation Matrix of characteristics on the dataset

the algorithm is developed, the next step is configuration and execution. This includes defining and adjusting the key parameters of FKE, such as the expansion rate, the balance between local and global exploration, and convergence criteria. The training phase involves applying the algorithm to a subset of data to fine-tune and optimize its performance. During this phase, iterative testing is necessary to improve the algorithm's effectiveness. Cross-validation is crucial for assessing the robustness of the model, ensuring that FKE can handle different datasets and maintain clustering accuracy.

4. **Evaluation of Results:** Evaluating the results is a key component for measuring the algorithm's success. Evaluation metrics such as the Silhouette index and Davies-Bouldin coefficient should be used to assess clustering quality. This analysis will help determine the effectiveness of FKE and qualitatively evaluate the model and its feasibility in this context.



1. **Data Extraction:** This step focuses on loading the necessary data for the analysis, in this case, from a file or an external source. The quality and initial structure of the data are essential to ensure proper processing in later stages.
2. **Data Validation:** At this stage, checks are performed to ensure that the data is complete and free of inconsistencies. This includes identifying null values, duplicates, or anomalies.
3. **Data Preparation:** In this step, data cleaning and feature selection are carried out. For this project, some irrelevant columns were removed, and the data was not standardized, as better metrics were obtained without this step.
4. **Model Training:** The bioinspired algorithm was applied using various distance metrics (Manhattan, Euclidean, Minkowski). The best performance was achieved with the Manhattan distance.
5. **Model Evaluation:** Metrics such as silhouette and Davies-Bouldin were used to measure the quality of the clusters. This step allowed us to identify that the metrics were consistent regardless of the algorithm's local and global radius.
6. **Model Validation:** It was confirmed that the results were stable when varying hyperparameters such as the number of clusters and the initial configuration.

6 Definition of the Model

The Fungi Kingdom Expansion (FKE) algorithm is based on the natural behavior of fungal colonies and their ability to adapt and grow in search of resources. This algorithm operates on two main phases: local expansion and global adjustment, which simulate the behavior of fungi as they explore and colonize their environment. Below is a detailed description of the model's characteristics and functionality.

1. **Local Expansion (Chaotic):** In the first phase, the model simulates the chaotic expansion of fungal spores. Each point in the dataset is considered a "nutrient patch," and the algorithm assigns these patches to the nearest fungal colonies (centroids). During this process, random noise is introduced into the distance calculations, simulating the unpredictable nature of fungal growth. This perturbation helps the model escape local optima and explore new areas within the data space, reflecting how fungi might disperse their spores in search of favorable conditions.
2. **Global Adjustment (Directed):** After the local expansion phase, the model enters the global adjustment phase. In this stage, the positions of the fungal colonies are recalculated based on the nutrient patches they are assigned to. The new centroid of each colony is determined by calculating the average positions of the assigned points, with a global adjustment that simulates how fungi relocate themselves toward rich nutrient sources. This balance between exploration (chaotic) and exploitation (directed) is fundamental to the algorithm's effectiveness.
3. **Convergence:** The algorithm checks for convergence of the fungal colonies by comparing the previous and current centroid positions. If the change in positions is less than a defined tolerance, the algorithm considers that convergence has been reached and stops.

7 Results

During the development of the project, the algorithm was tested using different distance functions and applying different feature selection techniques, using all the columns, eliminating features based on the analysis of the correlation matrix, and using a PCA with 98% of the variance explained, obtaining the best results by eliminating the following features: Purchases, MinimumPayments, CashAdvanceTrx, PurchasesInstallmentsFrequency, Tenure, PRC-FullPayment.

These characteristics were eliminated after performing the analysis of the correlation matrix, observing that some of them, such as Minimum Payments and Tenure, provide little relevant information; others, such as Purchases and PurchasesInstallmentsFrequency, have a high correlation with

other variables and therefore provide redundant information with other characteristics that favor the separation.

Table 2: Clustering Metrics Based on the Distance Function Used

Results		
Distance	Silhouette	Davies-Bouldin
Manhattan	0.49	1.28
Euclidean	0.39	1.30
Minkowski	0.31	1.36

Table 3: Clustering Metrics Based on the Number of Colonies

Results		
# Colonies	Silhouette	Davies-Bouldin
3	0.53	1.05
5	0.49	1.28
7	0.44	1.18
10	0.42	1.23

In Table 2 we can see the comparison of the metrics: Silhouette and Davies-Bouldin for the model with different distance functions, these tests were performed after the selection of characteristics through analysis of the correlation matrix, obtaining the best performance with the Manhattan distance. On the other hand, in Table 3 we can see the values of the same metrics for a different number of colonies (clusters), tests performed on the model with the Manhattan distance and the same data used for the previous table. The best performance is observed with 3 colonies, with the highest Silhouette value and the lowest Davies-Bouldin value, indicating a good quality of the clusters, with points similar to each other within a single cluster and a good separation between the different clusters.

In Figure 2 we can see a two-dimensional representation of the clusters generated by our algorithm made by PCA, this visualization allows us to get an idea of how the clusters are formed in our space, showing the similarity between the points belonging to each cluster.

However, as we originally had such a high dimensionality in our data (a total of 17 features) this representation does not allow us to clearly see the separation between the different clusters, having cluster 1 and 2 with a fuzzy boundary.

8 Conclusions

Although the algorithm presents a performance that can be considered acceptable, identifying dominant patterns allowing an adequate initial segmentation, the Silhouette and

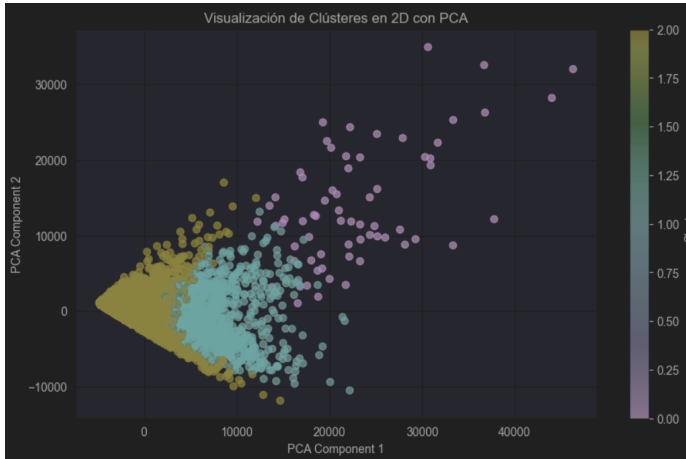


Figure 2: 2D PCA Visualization Clusters

Davies-Bouldin metrics indicate that the algorithm is capable of generating compact clusters and with a good separation, however the values of these indicate a wide margin for improvement, with clusters that are not perfectly defined.

In Figure 2 we can see a fuzzy boundary between the clusters, which may be mainly due to the chaotic behavior of our algorithm, which does not allow us to define perfect boundaries between the different fungal colonies, clusters in our case. This visualization allows us together with the value of the Silhouette and Davies-Bouldin metrics to observe the impact of the chaotic behavior typical of bio-inspired algorithms, presenting a challenge in parameter tuning to find the combination that best generates results for the specific problem we are working on.

This algorithm cannot be recommended over traditional approaches such as K-means or DBSCAN, however, some strengths of this algorithm can be highlighted, for example, it is able to capture nonlinear patterns thanks to its chaotic approach, being a bio-inspired algorithm faces complex distributions in a good way, since it has a high adaptive capacity. Precisely because of this chaotic approach and the randomness it entails, the algorithm is sensitive to noise and requires a careful adjustment of parameters, increasing the complexity and the time needed to obtain a good performance.

This algorithm has advantages and disadvantages with respect to other more deterministic clustering methods, being able to discover complex patterns but being sensitive to noise, with results more difficult to interpret, and a complexity susceptible to increase depending on the problem in which it is used. This is why its possible uses are reduced to exploratory and/or experimental applications due to its lack of reproducibility, stability and scalability.

9 References

- [Alnahwi et al., 2021] Alnahwi, F. M., Al-Yasir, Y. I. A., Sattar, D., Ali, R. S., See, C. H., and Abd-Alhameed, R. A. (2021). A new optimization algorithm based on the fungi kingdom expansion behavior for antenna applications. *Electronics*, 10(17):2057.
- [Barz et al., 2024] Barz, J. et al. (2024). Credit card dataset for clustering using k-means. *GitHub Repository*.
- [Bhasin, 2013] Bhasin, A. (2013). Credit card dataset for clustering. Accessed: September 2024.
- [de Ingeniería del Conocimiento, 2024] de Ingeniería del Conocimiento, I. (2024). Bioinspired algorithms for optimization and financial analysis. *IIC News*.
- [Fuchigami and Abreu, 2023] Fuchigami, H. Y. and Abreu, A. P. (2023). Innovative optimization algorithms for large-sized industrial scheduling problems. *SciELO*.
- [Huang et al., 2023] Huang, X., Liu, A., and Wang, J. (2023). Energy consumption optimization in data centers using fungi-inspired algorithms. *Journal of Sustainable Computing*.
- [Lin et al., 2021] Lin, S., Zhang, Y., and Zhao, H. (2021). Financial data analysis and consumer behavior using fungi kingdom expansion algorithm. *Journal of Financial Data Analytics*.
- [Unirioja, 2024] Unirioja, D. (2024). Applications of machine learning for credit card risk assessment. *Dialnet*.
- [Zhao et al., 2022] Zhao, Y., Shen, L., and Lü, Z. (2022). Fungi-inspired algorithms for smart manufacturing scheduling and resource optimization. *Journal of Industrial Optimization*.